

ADENTAN MUNICIPAL EDUCATION OFFICE

GovPay Desk

Individual contribution

Whitney Agyapomaa Bamfo

Student ID 22424615

CSIT622 Capstone Project · Group 7

University of Ghana · Department of Computer Science

Whitney Agyapomaa Bamfo: Approval routing and authority bands

I own approval routing: the rules deciding who may release a payment of a given size, and how a voucher reaches the right person.

The office's real constraint is that authority is bounded by amount. I modelled this as four bands (Routine up to GHS 5,000, Standard up to 50,000, Senior up to 250,000, and Executive without ceiling), held in one service rather than scattered through the interface. `ApprovalRouter::canApprove` is called from the voucher policy, so the same rule that greys out a button also refuses the request if someone bypasses the interface entirely. Hiding a control is a courtesy; the policy is the actual constraint.

The code this covers

- `app/Services/ApprovalRouter.php`
- `app/Services/Notifier.php`
- `app/Http/Controllers/NotificationController.php`

Questions to be ready for

The examiner may ask about any part of the system, not only this one. These are the ones closest to the work described above:

- What are the four bands, and what happens when a voucher exceeds the approver's ceiling?
- Why is `canApprove` called from the policy rather than checked in the controller?
- How does a voucher reach the right approver, and what happens if nobody at that level exists?

Before submitting, confirm this describes what you actually did. This section is assessed individually, and you may be asked to demonstrate any claim in it. Rewrite anything that does not match your own work, and add what is missing, particularly what you found difficult and what you would do differently now.